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Modulation of α-Synuclein Aggregation Amid Diverse Environmental Perturbation

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Abstract

Intrinsically disordered protein α-Synuclein (αS) is implicated in Parkinson's disease due to its aberrant aggregation propensity. In a bid to identify the traits of its aggregation, here we computationally simulate the multi-chain association process of αS in aqueous as well as under diverse environmental perturbations. In particular, the aggregation of αS in aqueous and varied environmental condition led to marked concentration differences within protein aggregates, resembling liquid-liquid phase separation (LLPS). Both saline and crowded settings enhanced the LLPS propensity. However, the surface tension of αS droplet responds differently to crowders (entropy-driven) and salt (enthalpy-driven). Conformational analysis reveals that the IDP chains would adopt extended conformations within aggregates and would maintain mutually perpendicular orientations to minimize inter-chain electrostatic repulsions. The droplet stability is found to stem from a diminished intra-chain interactions in the C-terminal regions of αS, fostering inter-chain residue-residue interactions. Intriguingly, a graph theory analysis identifies *small-world-like networks* within droplets across environmental conditions, suggesting the prevalence of a consensus interaction patterns among the chains. Together these findings suggest a delicate balance between molecular grammar and environment-dependent nuanced aggregation behaviour of αS.

eLife assessment

This study provides **important** biophysical insights into the molecular mechanism underlying the association of alpha-synuclein chains, which is essential for understanding the pathogenesis of Parkinson's disease. The data analysis is **solid**, and the methodology can help investigate other molecular processes involving intrinsically disordered proteins.

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Introduction

In the human body, a significant presence of Intrinsically Disordered Proteins (IDPs) plays diverse and crucial roles.[1–3] These proteins lack a well-defined 3D structure under native conditions, which imparts functional advantages, but also renders them susceptible to irreversible aggregation, especially when affected by mutations. Such aggregates can be pathogenic and are associated with various diseases, including neurodegenerative diseases, cancer, diabetes, and cardiovascular diseases.[4]

Notably, Alzheimer's disease is characterized by the aggregation of the amyloid-β peptide (Aβ), while Parkinson's disease (PD) is linked to α-Synuclein (αS) aggregation. A growing body of evidence has established a connection between IDPs and the phenomenon known as *liquid-liquid phase separation* (LLPS). During LLPS, high and low concentrations of biomolecules coexist without the presence of membranes and exhibit properties similar to phase-separated liquid droplets of two immiscible liquids (**Figure 1**).[5–9] This intriguing phenomenon has garnered significant attention as it underlies the formation of membrane-less subcellular compartments,[10–12] which, when dysregulated, can lead to incurable pathogenic diseases.

Recent findings have highlighted the capability of αS to undergo LLPS under physiological conditions, specifically when the protein concentration surpasses a critical threshold.[6] Moreover, it was observed that the aggregation propensity of αS is significantly influenced by various factors, including the presence of molecular crowders, the ionic strength of the protein environment, and pH.[13] Nonetheless, characterizing the interactions and dynamics of these small aggregates poses experimental challenges, leading to limited available reports on the subject.[14–17]

This investigation aims to establish the molecular basis of self-aggregation of αS and underlying process of LLPS under diverse environmental perturbations. In particular, to understand the influence of environmental factors on the inter-protein interactions within a phase separated droplet, we target to computationally simulate the aggregation process of αS under different conditions, emphasizing the roles of crowders and salt. While recent progress in computational forcefields and hardware has enabled the simulation of individual Intrinsically Disordered Proteins (IDPs) especially αS, using All-Atom Molecular Dynamics (AAMD),[18–24] these simulations can be extremely time-consuming and resource-intensive, making multi-chain AAMD simulations, even with cutting-edge software and hardware, impractical. Therefore, to simulate the the aggregation process, we resort to coarse-grained molecular dynamics (CGMD) simulations. Multiple coarse-grained forcefields have been developed with the sole purpose of fast and accurate simulations of IDPs and LLPS.[25–28] However these are implicit water, residue-level coarse-grained models. Therefore, here we leverage a tailored Martini 3 Coarse-Grained Force Field (CGFF)[29] for αS and use it to dissect the inter-protein interactions governing stable aggregate formation and LLPS. By leveraging the CGFF framework and building upon the groundwork laid by prior studies,[30–32] we have optimized water-protein interactions for αS. Our multi-chain microsecond-long CGMD simulations have resulted in comprehensive ensembles of significant protein aggregates spanning various scenarios.

As one of the key observations, our simulation unequivocally reveal LLPS-like attributes in the aggregates and show how these get modulated in presence of crowders and salt. The investigation unearths the intricate interplay of mechanical and thermodynamic forces in αS aggregation, achieved through meticulous data analyses. We elucidate the pivotal intra and inter-protein interactions governing LLPS-like protein droplet formation, unveiling the protein's primary sequence's role in aggregation. As would be shown in this article, a graph-based depiction of the droplet's architecture represents the proteins within droplets as constituting dense networks akin to *small-world networks*.

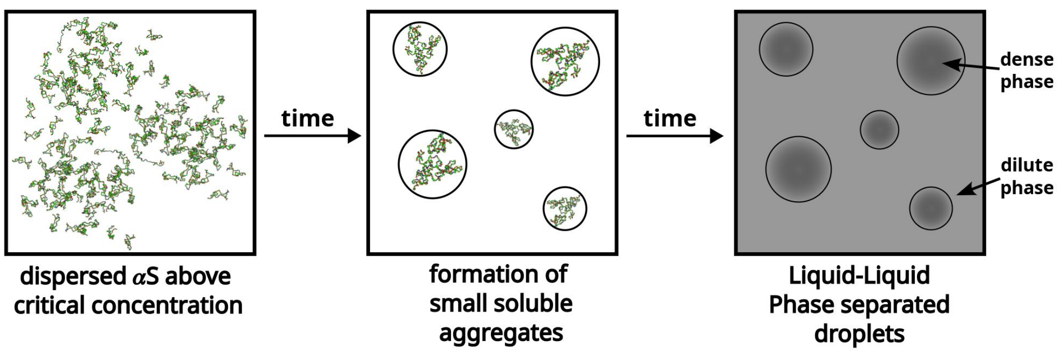

Figure 1.

A schematic showcasing the process of liquid-liquid phase separation of αS.

Results

In this study, we utilized the recently developed Martini 3[29] coarse-grained model to simulate collective interaction of a large number of αS chains in explicit presence of aqueous media at various concentrations commensurate with *in vitro* conditions including the presence of crowders and salt. As Martini 3 was not originally developed for intrinsically disordered proteins (IDPs), we carefully optimized the protein-water interactions against atomistic simulation of monomer and dimer of αS, as detailed in the *Methods* section, to ensure compatibility with αS (see *Methods*).

Initially, we examined the impact of concentration on the protein's aggregation by simulating copies of chains, maintaining a polydispersity of protein conformations of αS. In particular, three different conformations of αS (referred here as *ms*1, *ms*2 and *ms*3) with R_gs (radius of gyration) ranging between collapsed and extended states (1.84-5.72 nm) at different concentrations, with a composition, as estimated in a recent investigation,[23] were employed. First the chains were simulated for extensive period in a set of three protein concentrations, close to previous experiments.

Simulations capture enhanced aggregation beyond a threshold concentrations of αS

We performed simulations of αS at various concentrations, namely 300 μM, 400 μM, 500 μM and 750 μM. We begin by analysing the aggregation behavior of αS. As shown in **Figure 2**, we observe that most chains do not aggregate at 300 and 400 μM as characterized by the prevalence of high number of free monomers. The respective snapshots of the simulation indicate the presence of greater extent of single chains. Also, the chains that are not free, form very small oligomers of the order of dimer to tetramer (**Figure 2**).

However, upon increasing the concentration to 500 μM, which has also been the critical concentration reported for αS to undergo LLPS,[6] we observe a sharp drop in the average number of free monomers in the system (**Figure 2**). The corresponding representative snapshot of the system also depicts a few higher order aggregates, such as pentamers and hexamers, as well as most chains forming small oligomers. This can be understood from the value of the average number of chains present in the largest clusters, as reported in **Figure 2**.

The system, being at critical concentration, formation of large aggregates would require longer timescales than the simulation length. Therefore, in order to promote formation of large aggregates (heptamers or more) for finer characterization, we performed a simulation at a higher concentration of 750 μM αS. As shown in **Figure 2d**, we observe further decrease in the total number of free monomeric chains in the solution. There is simultaneous appearance of a very few droplet like aggregates (hexamer or more) as can be seen from **Figure 2** and the adjacent snapshot of the system (**Figure 2**). However, we note that ∼60 % of the protein chains are free and do not participate in aggregation and we think that as such in water, αS does not possess a strong and spontaneous self-aggregation tendency. In the following sections we characterize the aggregation tendency of αS in presence of certain environmental modulator that can shed more light on this hypothesis.

Molecular crowders and salt accelerate αS aggregation

The cellular environment, accommodating numerous biological macromolecules, poses a highly crowded space for proteins to fold and function.[33–36] In *in vitro* studies, inert polymers such as Dextran, Ficoll, and polyethylene glycol (PEG) are commonly employed as macro-molecular crowding agents. In the context of αS amyloid aggregation, previous experimental studies have

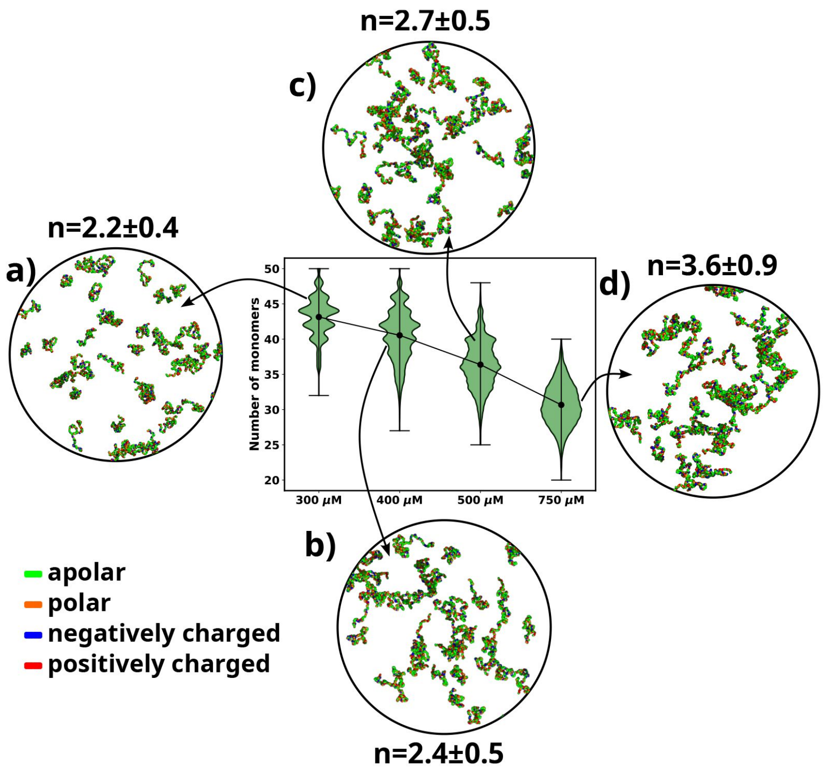

Figure 2.

A violinplot showing the distribution of number of monomers present for different concentrations of α-syn. The blue dot at the middle of each distribution represents the mean number of monomers observed for each concentration. For each concentration we show representative snapshots of the system. For each concentration, we also report the statistics of the number of chains in the largest cluster (n). **a)** A snapshot from the simulation at 300 μM α-syn. **b)** A snapshot from the simulation at 400 μM α-syn. **c)** A snapshot from the simulation for 500 μM α-syn. **d)** A snapshot from the simulation at 750 μM α-syn. Each chain in the snapshots has been coloured differently.

revealed an increased rate of *in vitro* fibrillation in the presence of different crowding agents.[37–39] Notably, a recent experimental study demonstrated the occurrence of phase separation (LLPS) of αS in the presence of PEG molecular crowder.[6] Moreover, considering that *in-vivo* environments also contain various moieties like salts and highly charged ions, a recent *in vitro* study has shown that the ionic strength of the solvent directly influences the aggregation rates of αS,[13] with higher ionic strength enhancing αS aggregation.

Given these observations, it becomes crucial to characterize the factors responsible for the enhanced aggregation of αS in the presence of crowders and salt. To address this, we perform two independent sets of simulations: one with αS present at 750 μM in the presence of 10% (v/v) fullerene-based crowders (see *SI Methods*) and the other with the same concentration of αS but in the presence of 50 mM of NaCl. In this section we characterize the effects of addition of crowders or salt on the aggregation of αS.

As expected, the addition of crowders leads to an enhancement of αS aggregation due to their volume excluded effects, as depicted in **Figure 3a**. Notably, the number of monomers drastically decreases upon the inclusion of crowders. This observation is further supported by the snapshots of the system, which also confirm the reduction in monomer count. Similarly, we observe that the presence of salt also promotes αS aggregation, as illustrated in **Figure 3a**, where the number of monomers is lower when compared to the case with no salt.

Following this, we conducted an analysis of the number of chains present in the largest clusters that formed. **Figure 3b** clearly illustrates that the addition of crowder or salt leads to a notable increase in the average number of proteins forming a cluster. This crucial observation points to the fact that the inclusion of accelerators, such as crowder or salt, not only promotes aggregation but also plays a role in stabilizing the formed oligomers. Importantly, we observed that the effect of crowder on aggregation is slightly more pronounced compared to that of the salt. In the subsequent section, we delve into the reasons behind the enhanced aggregation induced by these accelerators, aiming to decipher the underlying mechanisms responsible for their influence on αS aggregation dynamics. As the aggregation is significant enough for performing quantitative analysis only when the concentration of αS is 750 μM, we perform all analysis on scenarios at 750 μM of αS.

Crowders and salt differentially modulate surface tension for promoting LLPS-like αS droplets

The preceding sections underscore our simulation-based observation that, influenced by crowders and salt, αS aggregates into higher-order oligomers (hexamers and beyond) at a significantly accelerated propensity compared to the scenario without these influences. Here, we delve into the investigation of the energetic aspects underlying this aggregation phenomenon. An important contributor to the energetics is the surface tension, arising from the creation of interfaces between the dense and dilute phases of the protein upon droplet formation. This presence of interfaces is accompanied by surface tension and surface energy. The surface energy of a system is directly proportional to its surface area; systems with higher surface energy tend to minimize their surface area. Consequently, systems comprising multiple smaller droplets exhibit a larger surface area, and hence a higher surface energy. Conversely, systems characterized by fewer, larger droplets possess a comparatively reduced surface area and correspondingly lower surface energy. This insight leads us to conjecture that surface tension could play a pivotal role in driving liquid-liquid

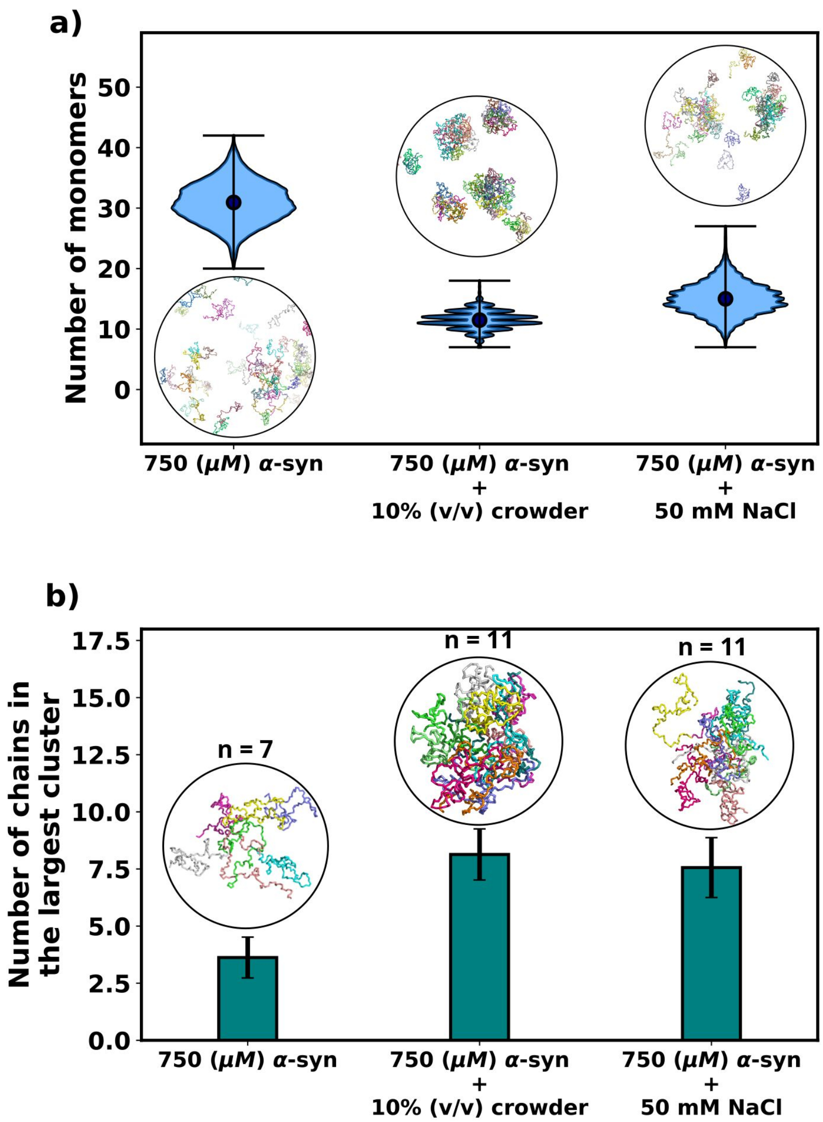

Figure 3.

a) A violinplot showing the distribution of the number of monomers for α-syn at 750 μM without and with crowder. The blue dots represent the means of each distribution. The snapshots represent the extent aggregation for a visual comparison. **b)** A bar plot showing the number of chain in the largest cluster formed by α-syn at 750 μM without and with crowder. The snapshots show the largest cluster formed for each scenario.

phase separation (LLPS) and the formation of larger αS droplets. To explore this hypothesis, we calculate the surface tension of the resultant droplets, as per Eqs-1, 2 and as described *SI Methods* and Ref.[30]

$$\gamma_{20} = \frac{5k_BT}{16\pi\langle(\delta a + \delta b)^2\rangle} \tag{1}$$

$$\gamma_{22} = \frac{15k_BT}{16\pi\langle(\delta a - \delta b)^2\rangle} \tag{2}$$

where $da = a - R$ and $db = b - R$ is the perturbation of the droplet shape from a perfect sphere with a radius R along any two pairs of principle axes of general ellipsoid estimating the shape of the droplet. The surface tension (γ) is thus estimated using $\gamma \approx \gamma_{20} \approx \gamma_{22}$. Please see *SI Methods* and Ref.[30] for more details.

Figure 4a provides a comparison of the surface tension (γ), for three different scenarios involving αS: i) αS in solution, ii) αS in the presence of crowders, and iii) αS in the presence of salt. Notably, in each case, the surface tension is considerably lower (0.0035-0.0075 mN/m) than the surface tension for FUS droplets in water (~0.05 mN/m).[30] As stated earlier, the magnitude of surface tension is an estimate of the aggregation tendency of any liquid-liquid mixture. Since we find that $\gamma_{\alpha S}$ is much lower than γ_{FUS}, we assert that the propensity with which αS aggregates should be much lower than that of FUS.

Next, we conduct a comparison of the three different scenarios to understand the effects of crowders and salt on the aggregation of αS. From **Figure 4a**, it is evident that the surface tensions are very similar for cases (i) and (ii), while it has increased for case (iii). This implies that the addition of crowders does not significantly impact the surface tension of the aggregates, although it renders the protein more prone to aggregation. On the other hand, the addition of salt causes an increase in surface tension. Given the relationship between surface area and volume, where a higher surface-to-volume ratio signifies numerous smaller droplets, the surface energy is concurrently elevated. In the presence of salt, a tendency is observed for these smaller aggregates to coalesce, giving rise to larger aggregates, albeit in reduced numbers. This behavior is an endeavor to curtail the surface-to-volume ratio and thus mitigate the associated surface energy. Therefore, the larger the surface tension, the higher is tendency of the protein to form aggregates, as seen from the surface tension values of αS and FUS, as mentioned earlier.

To minimize the surface energy, fusion of aggregates, either via merging of two or more droplets into one is seen for liquid-like phase separated droplets in experiments.[6] Although droplet fusion was not observed in our simulations due to the limited system size, it was shown that if a protein undergoes LLPS, a significant difference in protein concentration occurs between the droplet and the dilute phase.[40] To verify whether the aggregates observed in our simulations exhibit characteristics of LLPS, we calculated the protein concentrations in the dilute and concentrated phases. For the droplet phase, the concentration of the protein was calculated using Eq-3.

$$c_{phase} = \frac{N_{phase}}{N_A \times V_{phase}} \tag{3}$$

where N_{phase} is the number of protein chains in the phase (here dilute or concentrated), N_A is Avogadro's number and V_{phase} is the volume occupied by the phase. For the dilute phase, we estimated the volume of the concentrated/dense phase (V_{dense}) using Eq-4.[40]

$$V^i_{dense} = 4\pi\sqrt{3}\lambda_1\lambda_2\lambda_3 \tag{4}$$

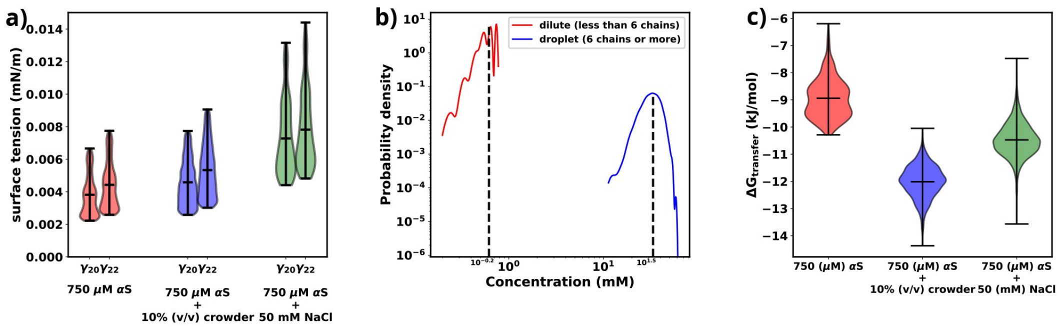

Figure 4.

a) surface tensions of droplets, estimated from γ_{20} and γ_{22}, for three cases has been shown. Both γ_{20} and γ_{22} provide almost similar estimates of the value of surface. **b)** Comparison of protein concentrations for the dilute(red) and the droplet (blue) phases. **c)** Excess free energy of transfer comparison for three cases.

where V_{dense}^{i} is the volume of the i-th droplet, λ_1, λ_2 and λ_3 are the eigenvalues of the gyration tensor for the aggregate. The volume of the dilute phase is the remainder volume of the system given by Eq-5.

$$V_{dilute} = V - \sum_{i} V_{dense}^{i} \tag{5}$$

where V is the total volume of the system.

As shown in **Figure 4b** and Figure S1, there is an almost two orders of magnitude difference between the concentration of αS in the dilute and droplet phases for all scenarios. Such a pronounced difference is a hallmark of LLPS, leading us to assert that the aggregates formed in our simulations possess LLPS-like properties. Consequently, we use the term "droplet" interchangeably with "aggregates" for the remainder of our investigation.

$$\Delta G_{transfer} = RT \ln\left(\frac{c_{dilute}}{c_{dense}}\right) \tag{6}$$

Finally, utilizing the calculated concentrations, we proceed to estimate the excess free energy of monomer transfer ($\Delta G_{transfer}$), from Eqn-6, between the dilute and droplet phases, where c_{dilute} is the concentration of αS in the dilute phase, c_{dense} is the concentration of αS in the dense/droplet phase, R is the universal gas constant and T is the temperature of the system (=310.15 K). As illustrated in **Figure 4c**, both crowder and salt scenarios demonstrate lower $\Delta G_{transfer}$ values compared to the case without their presence. However, the thermodynamic origins behind this pronounced aggregation differ for crowders and salt. Crowders enhance aggregation primarily through excluded volume interactions, which are of an entropic nature. On the other hand, salt enhances aggregation by increasing the droplet's surface tension, thus contributing to the enthalpy of the system. As a result, apart from the already known fact that macromolecular crowding decreases $\Delta G_{transfer}$ via entropic means, we also infer that salt decreases $\Delta G_{transfer}$ via enthalpic means by increasing the surface tension of the formed droplets.

Aggregation results in chain expansion and chain reorientation in *α*S

An indicative trait of molecules undergoing Liquid-Liquid Phase Separation (LLPS) is the adoption of extended conformations upon integration into a droplet structure. Given that the aggregates observed in our simulations exhibit a concentration disparity reminiscent of LLPS between the dilute and dense phases, we endeavored to validate the presence of a comparable chain extension phenomenon within our simulations.[40] To address this, we quantified the radius of gyration (R_g) for individual chains and classified them based on whether they were situated in the dilute or dense phase. The distribution of R_g values for each category is illustrated in **Figure 5a** and Figure S2. Remarkably, the distribution associated with the dense phase distinctly indicates that the protein assumes an extended conformation within this context. As elucidated earlier, this marked propensity for extended conformations aligns with a characteristic hallmark of LLPS as previously seen in experiments.[41]

Having observed the conformational alterations of αS during LLPS, our subsequent aim was to quantify the extent of these conformational changes in relation to their initial states (referred as '*ms*1', '*ms*2' or '*ms*3' in decreasing order of R_g [23]). To achieve this, we computed the Root Mean Square Deviation (RMSD) of each protein relative to its starting conformation. The resulting distributions were visually depicted using violin plots, featuring bold edges in **Figures 5b** and **c**. The protein ensemble was segregated into two categories: i) those from the dilute phase (**Figure 5b**), and ii) those from the dense phase (**Figure 5c**).

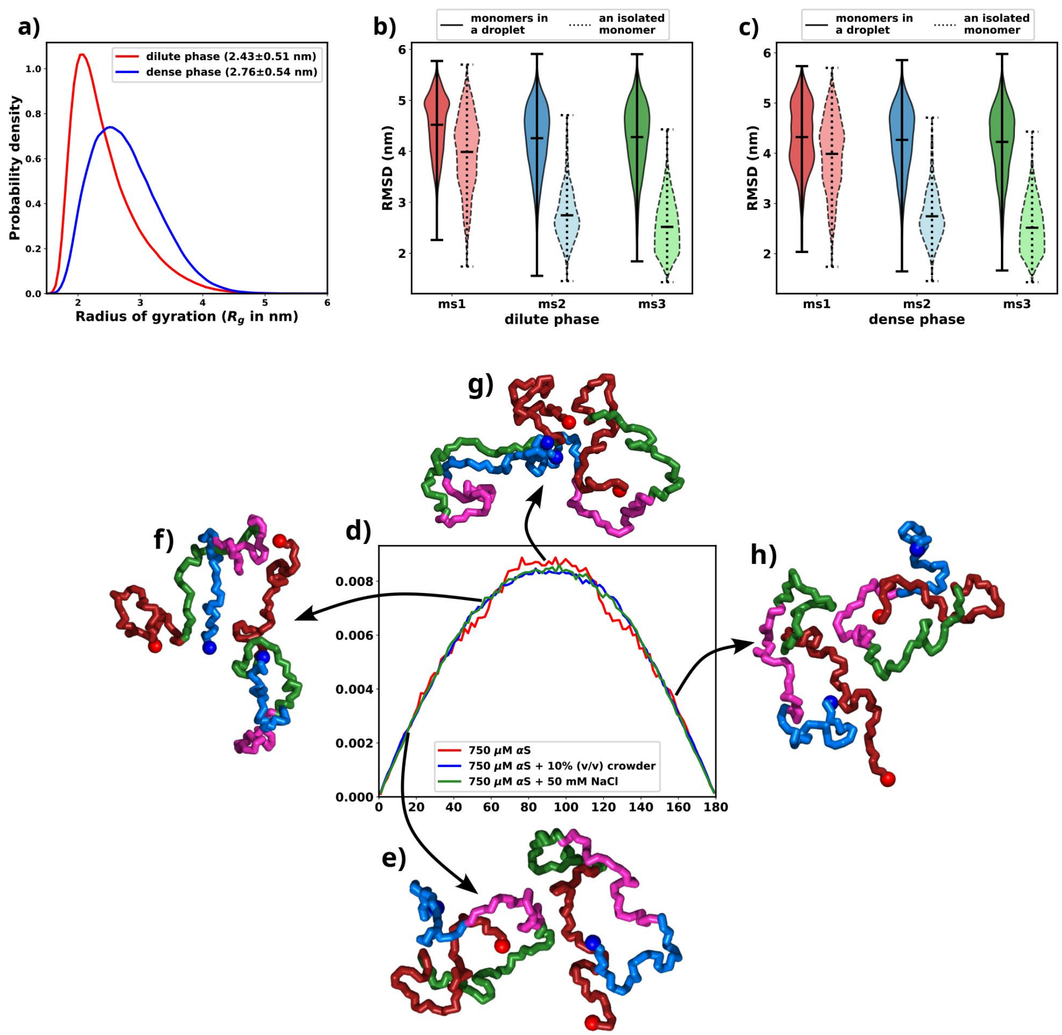

Figure 5.

All the figures are for 750 μM αS + 50 mM NaCl. **a)** Distribution of R_g for proteins present in the dense or the dilute phases. **b)** Comparison of RMSD for protein chains present in the dilute phase, with single chain RMSDs as the reference (dotted edges). **c)** Comparison of RMSD for protein chains present in the dense phase, with single chain RMSDs as the reference (dotted edges). **d)** Distribution of the angle of orientation of two chains inside the droplet for the three different scenarios. **e)** representative snapshot for angle between 0 degree and 20 degree. **f)** representative snapshot for angle between 50 degree and 70 degree. **g)** representative snapshot for angle between 80 degree and 120 degree. **h)** representative snapshot for angle between 150 degree and 180 degree.

Surprisingly, regardless of their initial configurations, the observed RMSD values were notably high. To facilitate a comparative analysis, we also included distributions of RMSDs for single chains simulated in the presence of 50 mM of salt, depicted using violin plots with broken edges. Intriguingly, the conformational state labeled as *ms*1, exhibited the least RMSD, a characteristic attributed to its notably extended conformation. This phenomenon aligns with the preference of droplets for extended conformations, implying that ms1 required the least conformational perturbation and thus exhibited a lower RMSD.

For both ms2 and ms3, a conspicuous increase in RMSD values was observed across all proteins monomers, irrespective of their respective phases. This phenomenon can potentially be attributed to the pronounced conformational shift experienced by the protein during aggregation. Building on these observations, we put forward a hypothesis: LLPS engenders significant modifications in the native protein conformations, ultimately favouring the adoption of extended states.

As discussed in the previous paragraph that the αS monomers inside the droplets must undergo conformational expansion and we hypothesized that they adopt orientation so as to minimize the inter-chain electrostatic repulsions. To this end, we try to decipher the orientations of the chains via defining their axes of orientations and subsequently calculating the angles between the major axis of two monomers. We calculate the major axis of gyration, given by the eigenvector corresponding to the largest eigenvalue of the gyration tensor, for each monomer inside a droplet. We next find the nearest neighbour (minimum distance of approach < 8 Å) for each monomer, carefully taking care of over-counting.

The angle between two monomers is defined as the angle between the major axes of gyration between chain i and its nearest neighbour j. We plot the distributions of the angles for all scenarios and all droplets in **Figure 5d**. We observe that irrespective of the conditions, the distribution peaks at right angles. The representative snapshots (**Figure 5e-5h**) showcase their mode of orientation. Interestingly the distribution is the same for all the three scenarios, again stressing upon the fact the αS droplets share similar features in terms of interactions and orientations irrespective of their environments.

Characterization of molecular interactions in aggregation-prone conditions

As established in preceding sections, both crowders and salt have been observed to augment the aggregation of αS while concurrently stabilizing the resultant aggregates. This phenomenon leads to the protein adopting extended conformations within a notably heterogeneous ensemble. Shifting our attention, we now delve into a residue-level investigation to unravel the specific interactions responsible for stabilizing these aggregates and, consequently, facilitating the aggregation process.

To compute the differential contact maps, our approach involved initial calculations of average intra-protein residue-wise contact maps, termed as intra-protein contact probability maps, for monomers present in both the dilute and dense phases (refer to Figures S3 and S4). Subsequently, we derived the difference by subtracting the contact probabilities of monomers within the dilute phase from those within the dense phase. As evident from **Figure 6a**, a discernible reduction in intra-chain Nter-Cter interactions is observed for monomers within the droplet phase, depicted by the presence of blue regions along the off-diagonals. Such a reduction in such interactions has also been observed via experiments[41] and it is similarly noticeable in the two other cases, as evident in **Figures 6b** and **6c**.

Furthermore, a significant decline in intra-protein interactions, especially the NAC-NAC interactions, is predominantly observed at shorter ranges, indicated by deep-blue regions concentrated near the diagonals. Notably, these diminished intra-chain interactions (**Figure 6**

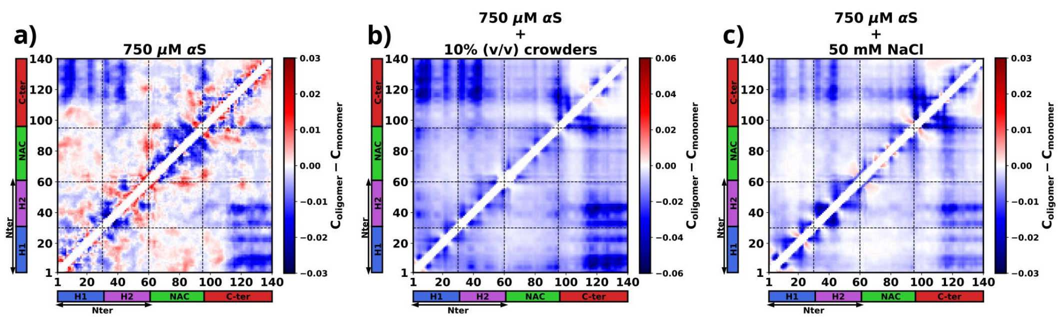

Figure 6.

The figure presents the residue-wise, intra-protein difference contact maps where the average contact probability of monomers in the dilute phase were subtracted from the average contact probability of monomers in the dense/droplet phase for three cases: **a)** 750 μM αS in water. **b)** 750 μM αS in the presence of 10% (v/v) crowders. **c)** 750 μM αS in presence of 50 mM NaCl.

and S3) potentially facilitate the formation of inter-chain interactions (Figure S4). Thus we observed that increased inter-chain NAC-NAC regions (Figure S5) facilitate the formation of αS droplets which also have been previously been seen from FRET experiments on αS LLPS droplets.[6] Building on these observations, we posit that these interactions play a pivotal role in stabilizing the aggregates that have formed.

Moreover, from the difference heatmaps in Figure S5, it can be observed that the residues 95-110 (VKKDQLGKNEEGAPQE) have reduced contact probabilities upon introduction of crowders/salt, whereas the rest of the contacts have slightly increased. These residues are highly charged and we think that upon introduction of crowders/salt, the proteins inside the droplet needed to be spatially oriented to facilitate the formation of largest aggregates. This re-orientation occurs to minimize the electrostatic repulsions among these residues belonging to different chains. These analyses provide hints that these residues are present in the protein so as to avoid the formation of aggregation prone conformations, which is why their interactions had to be minimized to form more stable and larger aggregates.

Phase separated *α*S monomer form *small-world* networks

The investigations so far suggest that irrespective of the factors that cause the aggregation of αS, the interactions that drive the formation of droplet remain essentially the same. However the conformations of the monomers vary depending on their environment. In presence of crowders they adapt to form much more compact aggregates. Therefore here we characterize whether the environment influences the connectivity among different chains of the protein inside a droplet.

Figure 7a, 7b and **7c** show molecular representations of the largest cluster formed by αS at 750 μM in water, αS at 750 μM in presence of 10% (v/v) crowders and αS at 750 μM in presence of 50 mM NaCl respectively. From the molecular representations for aggregates, it can be seen that irrespective of the system, they form a dense network whose characterization is not possible directly. Therefore we represent each aggregate as a graph with multiple nodes (vertices) and connections (edges), as can be seen from **Figure 7d, 7e** and **7f**. Each node (in blue) represents a monomer in the droplet. Two nodes have an edge (line connecting two nodes) if the minimum distance of approach of the monomers corresponding to the pair of nodes is at least 8 Å. We can see from the graph that not all chains are in contact with each other. They rather form a relay where a few monomers connect (interact) with most of the other protein chains. The rest of the chains have indirect connections via those. Since inter-chain connections/interactions have been denoted by edges and the chains themselves as nodes, such form of inter-chain interactions inside a droplet lead to only a few nodes having a lot of edges, for example node 1 in **Figure 7f**. The rest of them have only a few (3-5) edges. This is a signature of *small-world networks* [42–45] and we assert that αS inside the droplet(s) form small-world-like networks.

A network can be classified as a small-world network by calculating the *Clustering coefficient* and the average shortest path length for the network and comparing those to an equivalent *Erdos-Renyi* network.[46] The *clustering coefficient* (C) is a measure of the “connectedness” of a graph, indicating the extent to which nodes tend to cluster together. It quantifies the likelihood that two nodes with a common neighbor are also connected. On the other hand, the *average shortest path length* (L) is a metric that calculates the average number of steps required to traverse from one node to another within a network. It provides a measure of the efficiency of information or influence propagation across the graph. To estimate the *small-worldness* of a graph, we calculate a parameter (*S*) defined by Eq-7.

$$S = \frac{\frac{C}{C_r}}{\frac{L}{L_r}} \tag{7}$$

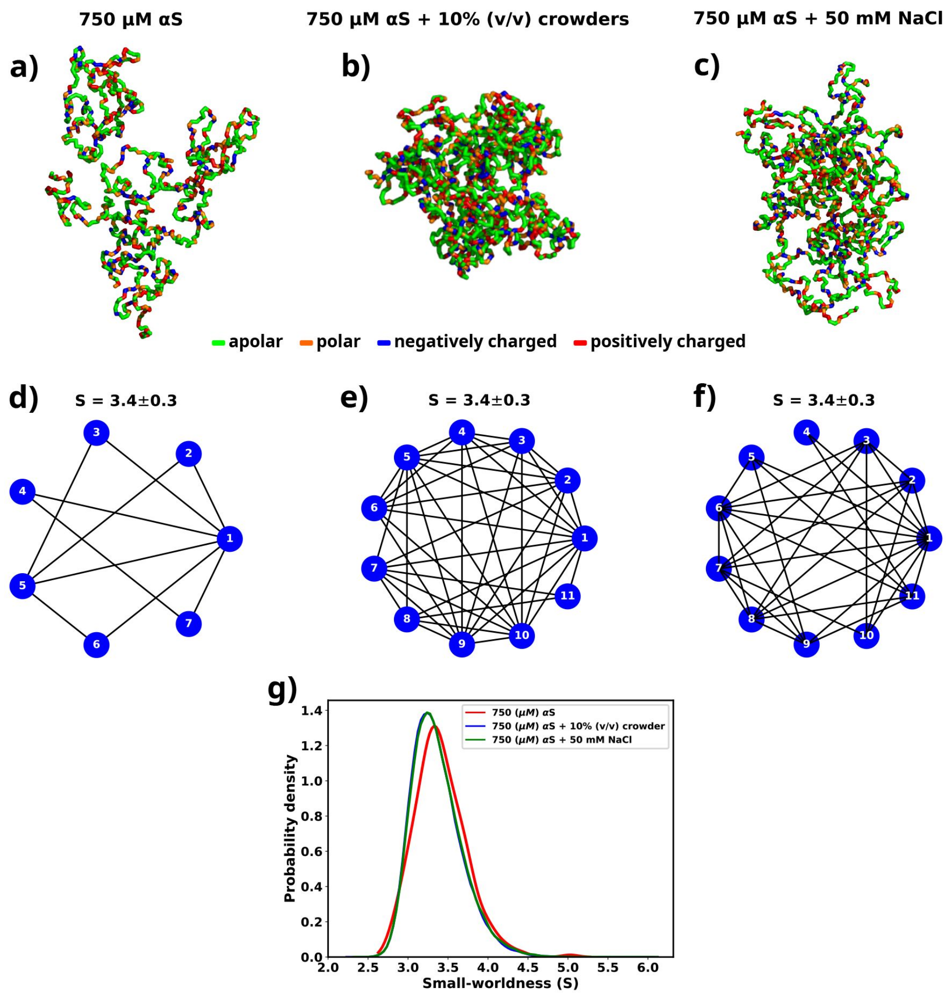

Figure 7.

a) The largest cluster formed by αS at 750 μM. **b)** The largest cluster formed by αS at 750 μM in the presence of 10% (v/v) crowder. **c)** The largest cluster formed by αS at 750 μM in the presence of 50 mM salt. Different residues have been color coded as per the figure legend. **d)** A graph showing the contacts among different chains constituting the largest cluster formed by αS at 750 μM. **e)** A graph showing the contacts among different chains constituting the largest cluster formed by αS at 750 μM in the presence of 10% (v/v) crowder. **f)** A graph showing the contacts among different chains constituting the largest cluster formed by αS at 750 μM in the presence of 50 mM NaCl. The mean small-worldness (S) of all droplet has been reported above the graph. **g)** Distribution of small-worldness (S) for all scenarios.

where C and L are the clustering coefficient and average shortest path length for the graph generated for a droplet while C_r and L_r clustering coefficient and average shortest path length for an equivalent Erdos–Renyi network. Small-world networks exhibit the characteristic property of having $C >> C_r$, while $L \approx L_r$. In light of this, for every scenario (solely αS, αS in the presence of crowder, and αS in the presence of salt), we generate an ensemble of graphs that correspond to the droplets formed during the simulation.

For each graph, we calculate the small-worldness coefficient (S)[44] and illustrate the distribution in **Figure 7g**. We observe a narrow distribution of S with a mean of 3.4 for all cases. In a previous report of RNA-LLPS, a value of $S \approx 4$ was used to classify the droplets small-world networks.[40] Therefore, $S = 3.4$ would suggest that the droplets formed during the simulations are small-world like. Moreover, we observe that the distribution of S is invariant with respect to the environment of the droplet.

Therefore we establish that the modes of interactions, orientations and even connectivities among αS monomers inside a droplet remain same even when their environments are extremely different. We think that this occurs since the residue-level interactions among different monomers inside the droplet are similar irrespective of the environment, as shown in a previous section. This puts forth a very interesting way of viewing αS LLPS. We think that if these residue-level interactions can be disturbed then the stability of the formed droplets might be affected in such a way that they might dissolve spontaneously.

Discussion

We used simulations to investigate the molecular basis of αS monomeric aggregation into soluble oligomers resembling micro-LLPS. The WT protein demonstrated limited aggregation, suggesting a low inherent propensity for LLPS dictated by its primary sequence. IDPs, like αS, often share primary sequence characteristics associated with phase separation. Charged residues distributed with uncharged amino acids, resembling the “sticker and spacer” model, contribute to this molecular grammar. This observation aligns with a general trend in IDPs.[47–49] To assess αS LLPS propensity from its primary sequence, we calculated Shannon entropy (S)[50] (Eq 8 and Table S2), Kyte-Doolittle hydrophobicity[51] (Table S3), Normalized, maximum of the sum of PLAAC Log-Likelihood Ratios (NLLR)[52] (Table S4), and LLPS propensity scores obtained from catGranules web-server[53] (Table S5).

$$S = \sum_i p_i \log p_i \tag{8}$$

where p_i is the probability of occurance of a residue in a given sequence.

Comparative analysis with three datasets,[54] namely LLPS+: a dataset of high propensity IDPs whose critical concentrations are 100 μM or below, LLPS-: a dataset of low propensity IDPs whose critical concentrations are greater than 100 μM, and PDB*: a dataset of folded proteins that do not undergo LLPS under normal conditions, revealed αS’s distinctive features (Table S6).

We note a significant difference in the Shannon entropy value of αS compared to proteins that do not undergo phase separation, as illustrated in **Figure 8a**. This deviation suggests a notable inclination of αS to undergo phase separation.[54] Additionally, the hydrophobicity of αS (**Figure 8b**) is lower than that of the PDB* dataset, aligning more closely with the upper extremes of the LLPS-dataset. This indicates that while αS exhibits a tendency to undergo phase separation, the propensity should be low. Consistent with this, NLLR scores obtained from PLAAC and LLPS propensity scores (**Figures 8c** and **d**) reinforce this observation. These collective comparisons, coupled with simulations and experimental data on its critical concentration,[6] conclusively establish that αS does not possess a high LLPS-forming propensity. Instead, this behavior is inherent to its primary structure. In hindsights, this analysis also justifies the requirements of environmental factors for enhancing the proclivity of αS for LLPS, as demonstrated in both our simulations and experimental findings.[6,13]

For characterizing αS's aggregation phenomena, we calculated droplet surface tension under varied conditions. We observed that crowders minimally impacted surface tension, while salt increased it; however, both scenarios decreased the relative free energy of the system. Crowders achieved this via entropic means, whereas salt employed enthalpic means. Residue-residue interactions during droplet formation were consistent across environments, with crowder or salt enhancing these interactions. The aggregation pathway involved overall inter-chain interaction enhancement, specifically reducing intra-chain Nter-Cter and Nter-NAC interactions, leading to more extended protein conformations in droplets. The comparison with reported FRET observations[6] aligns well with the findings from our simulations, indicating that within the droplets, intra-chain NAC-NAC interactions have been supplanted by inter-chain NAC-NAC interactions. Droplet proteins displayed consistent orientation and "small-worldness", a measure of inter-chain connectivity, remained consistent across diverse conditions. Thus, αS aggregates appeared invariant regarding their initial environment in terms of interactions and contacts.

Our study's precision was notably influenced by the careful selection of a simulation force-field. Despite the availability of modern force-fields optimized for multi-chain simulations of IDPs,[25,26,55,56] we opted for Martini 3, an explicit water model, due to its emphasis on water's role in aggregation and LLPS, as recently demonstrated in FUS LLPS.[57] Although newer models operate at a faster pace, Martini 3's inclusion of explicit water enhances result accuracy. Additionally, Martini 3 provides a detailed amino acid description and allowing for encoding of protein secondary structures, unlike some newer models that represent amino acids as single beads. Our meticulous choice of the simulation model, combined with a comprehensive analysis, contributes to the accuracy and novelty of this study.

Recent studies have explored the aggregation and LLPS of biopolymers and polyelectrolytes in the presence of membranes, opening a promising avenue for αS research.[58–60] Given that under physiological conditions, αS assumes an oligomeric, membrane-bound form, investigating its interactions with membranes could hold therapeutic potential.[61]

Under physiological conditions, crowding effects emerge prominently. While crowders are commonly perceived to be inert, as has been considered in this investigation, the morphology, dimensions, and chemical interactions of crowding agents with αS in both dilute and dense phases may potentially exert considerable influence on its LLPS. Hence, a comprehensive understanding through systematic exploration is another avenue that warrants extensive investigation.

Although we exclusively focused on wild-type αS, familial mutations have been reported to exhibit a significantly higher propensity for aggregation.[6] These mutations, involving minor alterations in the primary sequence, highlight the importance of understanding the molecular basis of this distinctive phenotype. Additionally, the observed stability of preformed αS droplets[62] poses a challenge in treating Parkinson's Disease (PD). Reversing aggregation/LLPS and understanding associated pathways and mechanisms are crucial. Our study identifies key residues crucial for stable droplet formation, consistent across various environmental conditions.

The significance of the solvent in αS aggregation remains underexplored. Recent studies[30,57] underscore the pivotal role of water as a solvent in LLPS. It suggests that comprehending the solvent's role, particularly water, is essential for attaining a deeper grasp of the thermodynamic and physical aspects of αS LLPS and aggregation. By delving into the solvent's contribution, researchers can uncover additional factors influencing αS aggregation. Such insights hold the potential to advance our comprehension of protein aggregation phenomena, crucial for devising

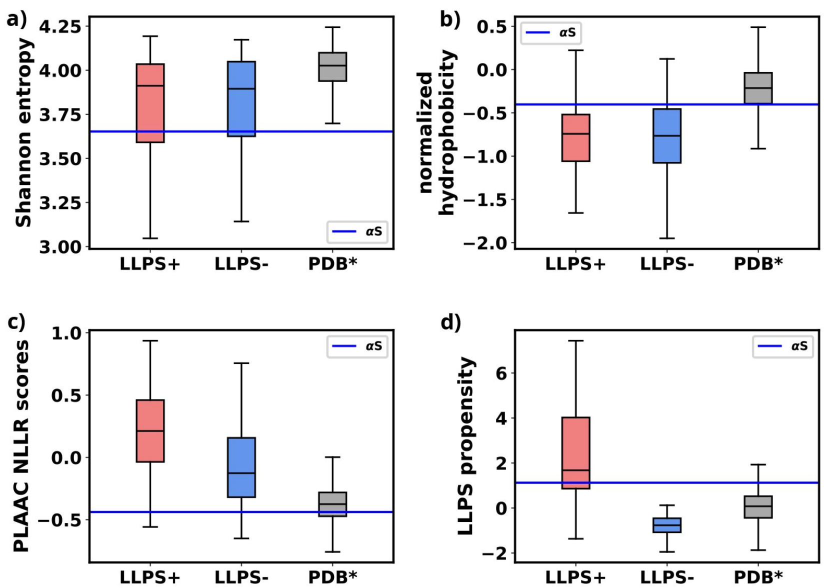

Figure 8.

a) Comparison of Shannon Entropy of different datasets with *α*S. **b)** Comparison of Kyte-Doolittle hydrophobicity of different datasets with *α*S. **c)** Comparison of LLR scores, obtained from PLAAC, of different datasets with *α*S. **d)** Comparison of LLPS propensity scores, obtained from catGRANULE websever, of different datasets with *α*S. The values have been summarized in Table S6.

strategies to address diseases linked to protein misfolding and aggregation, notably Parkinson's disease. Future investigations focusing on elucidating the interplay between αS, solvent (especially water), and other environmental elements could yield valuable insights into the mechanisms underlying LLPS and aggregation. Ultimately, this could aid in the development of therapeutic interventions or preventive measures for Parkinson's and related diseases.

Methods

Selection of the metric for optimizing water-protein interactions

We have opted to utilize the radius of gyration (Rg) of αS as the primary metric for optimizing water-protein interactions in Martini 3 for αS. To calibrate the Martini 3 force field, we employed 73 μs of all-atom data obtained from DE Shaw Research. From a polymer physics perspective, modifying water-protein interactions entails altering the solvent characteristics surrounding the biopolymer. We believe that Rg serves as an effective metric in this context. Additionally, we focus on matching the distribution of Rg values rather than solely targeting the mean value. This approach implies that, at a molecular level, the CGMD simulations conducted with optimized water-protein interactions enable the protein to explore conformations present in the all-atom ensemble.

Furthermore, we conducted cross-validation by comparing the fraction of bound states in all-atom and CGMD dimer simulations. This we claim that R_g is good metric to be used for tuning of water-protein interactions in Martini 3.

Optimizing Martini 3 parameters for αS

Martini 3[29] was trained using DES-Amber[63] that is an atomistic forcefield tuned for single-domain and multi-domain proteins. Therefore, the default parameters of the coarse-grained model is not suited for simulations of disordered proteins and reported to underestimate the global dimensions of these systems in addition to overestimating protein-protein interactions. Previous attempts to simulate IDPs have modified the Martini force field by tuning the water-protein interactions, specifically, σ and ϵ of Lennard-Jones interactions to render them suitable for modeling a specific IDP or all IDPs.[30,31,64] Here, we follow a similar protocol, however instead of tuning only the ϵ part of the water-protein Lennard-Jones interactions, we refine both the σ and ϵ parameters of the water-protein interactions (Eq-9).

$$V^{'}(r) = 4\epsilon^{'}\left[(\frac{\sigma^{'}}{r})^{12} - (\frac{\sigma^{'}}{r})^{6}\right] \tag{9}$$

where $\epsilon^{'} = \lambda\epsilon$, $\sigma^{'} = \lambda\sigma$ and λ is the scaling parameter that needs to be optimized. Scaling σ tunes the relative radius of the hydration spheres of each residue of a protein while a change in ϵ changes the strength of the water-residue interactions (**Figure 9a**). Increasing the ϵ value of water-protein interactions results in a higher energy demand for removing water molecules (dehydration) as a chain transitions from the dilute to the dense phase. Conversely, a higher σ value implies that the hydration shell will be at a greater distance, facilitating dehydration if a chain moves into the dilute phase. Therefore, adjusting water-protein interactions based on the protein's single-chain behavior may not significantly influence the protein's phase behavior. Furthermore, fine-tuning both ϵ and σ parameters only requires a minimal change in the overall protein-water interaction (1%). As a result, this adjustment minimally alters the force field parameters.

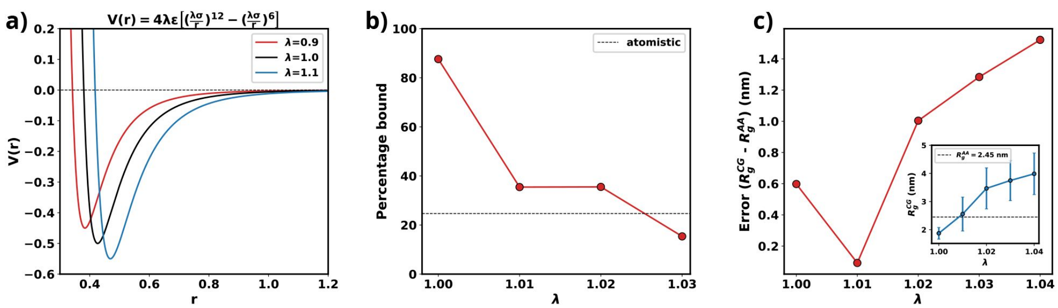

Figure 9.

a) Plot of LJ potentials with respect to λ. **b)** The percentage bound values between two CG αS chains for different values of λ. The dashed black line represents the percentage bound values for two all-atom chains. **c)** Error between R_g calculated from CG and from all-atom simulations vs λ. The inset plot showcases the average values of R_g obtained from CG along with their respective standard deviations. The dashed line represents the average value from all-atom simulations.

As we are interested in exploiting multi-chain simulations to study the LLPS of αS using Martini 3, we use the percentage of time two all-atom monomers remain bound to each other as the benchmark. To obtain an optimum scaling parameter for the water-protein interactions in Martini 3, specific to αS, we perform CG simulations with two αS chains with different values of λ. We start with two chains, without any secondary structure enforced upon them, randomly placed in a 15.7 nm box making sure that they are apart by at least 0.8 nm which we use as cutoff to classify the chains to be bound. If the minimum distance between any two residues belonging to the different chains are closer than 0.8 nm we consider them to be bound. Using the cutoff defined, we calculate the percentage bound between the two αS monomers for different values of λ in the coarse-grained model. We also calculate the same from atomistic simulations reported in[23] as the reference. From **Figure 9b**, we can see that for multiple values of λ, we observe a close agreement in percentage bound values between coarse-grained and atomistic simulations.

We conducted additional single-chain coarse-grained (CG) simulations of αS, varying the parameter λ, while refraining from imposing any secondary structure constraints. Subsequently, we compared the mean R_g values derived from these CG simulations with the 73 μs all-atom (AA) trajectory, which replaced the previously published 30 μs all-atom trajectory in[20] and was provided by DE Shaw Research. **Figure 9c** illustrates that, for λ = 1.01, the average R_g in the CG simulations closely matches the R_g values obtained from the all-atom data. Consequently, we have chosen λ = 1.01 for the multi-chain simulations, as it minimizes errors for both single-chain R_g and the observed percentage of time bound in the two-protein chain simulations.

Initial conformation generation for large-scale multi-chain simulations

A recent study used Markov State models to delineate the metastable states based on the extent of compaction (R_g) and identified 3 macrostates and their relative populations.[23] Subsequent to the investigation, we utilize three representative conformations, each corresponding to one of the macrostates. We designate these macrostates as 1 (ms1), 2 (ms2), and 3 (ms3) (refer to Figure S6). Therefore, in the multi-chain simulations, we maintain similar relative populations of these macrostates. (Figure S6). The reported percentages of macrostates (labeled as ms1, ms2 and ms3) are 0.06%, 85.9% and 14%, respectively. We added 50 αS monomers consisting of 1 chain of ms1, 45 chains of ms2 and 4 chains of ms3 in a cubic box with their respective secondary structures, determined via DSSP,[65–67] enforced using Martini 3. The size of the box of side a is determined as per Eq-10.

$$a = \sqrt[3]{\frac{N}{N_A \times C}} \tag{10}$$

where N is the number of monomers, N_A is the Avogadro's number and C is the required concentration of αS.

Here, we simulate multiple concentrations of the protein, namely, 300 μM, 400 μM, 500 μM and 750 μM. 300 and 400 μM are below the critical concentrations required to undergo LLPS.[6] We then solvate the system in coarse-grained water. We setup the 50-chain system to simulate 3 conditions; (a) in pure water, (b) in 50 mM NaCl and (c) in presence of 10 % (v/v) crowders. To study effect of salt, we add the required number of Na^+ and Cl^- ions to attain the desired concentration of 50 mM while also adding a few ions to render the system electrically neutral. In the system with crowders, we first add crowders after solvation by replacing a few solvent molecules with the required number of crowder molecules. We next resolvate the system along with the crowders. Finally we render the system electro-neutral by addition of the required number of Na^+ or Cl^- ions. The details of the simulation setup are provided in **Table 1**.

summary	conc. of αS(μM)	box size (nm)	# water	# crowders	# Na^+	# Cl^-
300 μM αS in water	300	65.66	2304122	0	450	0
400 μM αS in water	400	59.69	1729213	0	450	0
500 μM αS in water	500	55.52	1389721	0	450	0
750 μM αS in water	750	48.42	920023	0	450	0
750 μM αS + 10%(v/v) crowder	750	48.42	843011	20128	450	0
750 μM αS + 50 mM NaCl	750	48.42	913357	0	3783	3333

*For all simulations, a total of 50 monomeric protein chains have been used which comprise of $1\times MS1$, $45\times MS2$ and $4\times MS3$.

Table 1

Details of the systems that were explored

Simulation setup

Upon successful generation of the initial conformation, we first perform an energy minimization using steepest gradient descent using an energy tolerance of 10 kJ mol^{-1} nm^{-1}. We next perform NVT simulations at 310.15 K using v-rescale thermostat for 5 ns using 0.01 ps as the time step. It is then followed by NPT simulation at 310.15 K and 1 bar using v-rescale thermostat and Berendsen barostat for 5 ns with a time step of 0.02 ps.

Next we perform CG MD simulations using velocity-verlet integrator with a time step of 0.02 ps using v-rescale thermostat at 310.15 K and berendsen barostat at 1 bar. Both Lennard-Jones and electrostatic interactions are cutoff at 1.1 nm. Coulombic interactions are calculated using Reaction-Field algorithm and relative dielectric constant of 15. We perform CG MD for at least 2.5 μs for the systems with 50 αS monomers. The details of the simulation run-times have been provided in Table S1. We use the last 1 μs for further analyses.

Ascertaining the attainment of steady state in simulation

In this study, we utilized the final 1 μs from each simulation for further analysis. To ascertain whether the simulations have achieved a steady state, we plotted the time profile of protein concentration in the dilute phase for all three cases.

Except for minor intermittent fluctuation involving only αS in neat water (Figures S7a and S7b), the remaining cases exhibit notably stable concentrations throughout various segments of the trajectory (Figures S7c-f). The relatively higher fluctuations observed in Figures S8a and b stem from the slow, spontaneous aggregation of αS alone, compounded by the inherently ambiguous nature of the dense phase. Consequently, the addition or removal of a few chains from the dense to the dilute phase results in significant fluctuations in protein concentration within the dilute phase. Conversely, in the other two scenarios (Figures S7c-f), aggregation is expedited by the presence of crowders/salt, leading to the formation of larger aggregates. Consequently, the addition or removal of one or two chains has negligible impact on concentration, thereby mitigating sudden large jumps. In summary, the conspicuous jumps depicted in Figures S8a and b arise from the gradual, fluctuating aggregation of pure αS and finite size effects. However, since these remain within the realm of fluctuations, we assert that the systems have indeed reached steady states. This assertion is bolstered by the subsequent figure, where the time profile of several pertinent system-wide macroscopic properties reveals no discernible change between 1.5-2.5 μs (Figures S8).

List of Software

We have used only open-source software for this study. All simulations have been performed using GROMACS-2021.[68,69] Snapshots were generated using PyMOL 2.5.4.[70–72] Analysis were performed using Python[73] and MDAnalysis.[74,75] Figures were prepared using Matplotlib,[76] Jupyter[77] and Inkscape.[78]

Data Availability Statement

All data are present within the manuscript. The sections of the trajectories used for analysis (1.5 - 2.0 μs) have been uploaded to zenodo (*https://zenodo.org/records/10926368*). Other relevant files related to the simulations have also been uploaded to the same repository. Additional data can be made available upon request.

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(78) Inkscape Project **Inkscape Project. Inkscape.**

Editors

Reviewing Editor
Bin Zhang
Massachusetts Institute of Technology, Cambridge, United States of America

Senior Editor
Qiang Cui
Boston University, Boston, United States of America

Reviewer #1 (Public Review):

Summary:

In this paper, the authors performed molecular dynamics (MD) simulations to investigate the molecular basis of association of alpha-synuclein chains under molecular crowding and salt conditions. Aggregation of alpha-synuclein is linked to the pathogenesis of Parkinson's disease, and the liquid-liquid phase separation (LLPS) is considered to play an important role in the nucleation step of the alpha-synuclein aggregation. This paper re-tuned the Martini3 coarse-grained force field parameters, which allows long-timescale MD simulations of intrinsically disordered proteins with explicit solvent under diverse environmental perturbation. Their MD simulations showed that alpha-synuclein does not have a high LLPS-forming propensity, but the molecular crowding and salt addition tend to enhance the tendency of droplet formation and therefore modulate the alpha-synuclein aggregation. The MD simulation results also revealed important intra and inter-molecule conformational features of the alpha-synuclein chains in the formed droplets and the key interactions responsible for the stability of the droplets. These MD simulation data add biophysical insights into the molecular mechanism underlying the association of alpha-synuclein chains, which may be useful for understanding the pathogenesis of Parkinson's disease.

Strengths:

(1) The re-parameterized Martini 3 coarse-grained force field enables the large-scale MD simulations of the intrinsically disordered proteins with explicit solvent, which will be useful for a more realistic description of the molecular basis of LLPS.

(2) This paper showed that the molecular crowding and salt contribute to the modulation of the LLPS through different means. The molecular crowding minimally affects surface tension, but adding salt increases surface tension. It is also interesting to show that the aggregation pathway involves the disruption of the intra-chain interactions arising from C-terminal regions, which potentially facilitates the formation of inter-chain interactions.

Weaknesses:

(1) Although the authors emphasized the advantage of the Martini3 force field for its explicit description of solvent, this paper did not analyze the water behavior contained in the simulation trajectories and discuss the water's role in the aggregation and LLPS.

(2) This paper discussed the effects of crowders and salt on the surface tension of the droplets. The calculation of the surface tension relies on the droplet shape. However, for the formed clusters in the MD simulations, the typical size is <10, which may be too small to rigorously define the droplet shape. As shown in previous work cited by this paper [Benayad et al., J. Chem. Theory Comput. 2021, 17, 525−537], the calculated surface tension becomes stable when the chain number is larger than 100.

(3) Both the sizes and volume fractions of the crowders can affect the protein association. It will be interesting to perform MD simulations by adding crowders with various sizes and volume fractions. In addition, in this work the crowders were modelled by fullerenes, which contribute to protein aggregation mainly by entropic means as discussed in the manuscript. It is not very clear how the crowder effect is sensitive to the chemical nature of the crowders (e.g., inert crowers with excluded volume effect or crowders with non-specific attractive interactions with proteins, etc).

https://doi.org/10.7554/eLife.95180.2.sa2

Reviewer #2 (Public Review):

In the manuscript "Modulation of α-Synuclein Aggregation Amid Diverse Environmental Perturbation", Wasim et al describe coarse-grained molecular dynamics (cgMD) simulations of α-Synuclein (aSyn) at several concentrations and in the presence of molecular crowding agents or high salt. They begin by bench-marking their cgMD against all-atom simulations by Shaw. They then carry 2.4-4.3 μs cgMD simulations under the above-noted conditions and analyze the data in terms of protein structure, interaction network analysis, and extrapolated fluid mechanics properties. This is an interesting study because a molecular scale understanding of protein droplets is currently lacking.

https://doi.org/10.7554/eLife.95180.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1***
>
> *Summary:*
>
> *In this paper, the authors performed molecular dynamics (MD) simulations to investigate the molecular basis of the association of alpha-synuclein chains under molecular crowding and salt conditions. Aggregation of alpha-synuclein is linked to the pathogenesis of Parkinson's disease, and the liquid-liquid phase separation (LLPS) is*

> *considered to play an important role in the nucleation step of the alpha-synuclein aggregation. This paper re-tuned the Martini3 coarse-grained force field parameters, which allows long-timescale MD simulations of intrinsically disordered proteins with explicit solvent under diverse environmental perturbation. Their MD simulations showed that alpha-synuclein does not have a high LLPS-forming propensity, but the molecular crowding and salt addition tend to enhance the tendency of droplet formation and therefore modulate the alpha-synuclein aggregation. The MD simulation results also revealed important intra- and inter-molecule conformational features of the alpha-synuclein chains in the formed droplets and the key interactions responsible for the stability of the droplets. These MD simulation data add biophysical insights into the molecular mechanism underlying the association of alpha-synuclein chains, which is important for understanding the pathogenesis of Parkinson's disease.*
>
> *Strengths:*
>
> *(1) The re-parameterized Martini 3 coarse-grained force field enables the large-scale MD simulations of the intrinsically disordered proteins with explicit solvent, which will be useful for a more realistic description of the molecular basis of LLPS.*
>
> *(2) This paper showed that molecular crowding and salt contribute to the modulation of the LLPS through different means. The molecular crowding minimally affects surface tension, but adding salt increases surface tension. It is also interesting to show that the aggregation pathway involves the disruption of the intra-chain interactions arising from C-terminal regions, which potentially facilitates the formation of inter-chain interactions.*

We thank the reviewer for pointing out the strengths of our study.

> *Weaknesses:*
>
> *(1) Although the authors emphasized the advantage of the Martini3 force field for its explicit description of solvent, the whole paper did not discuss the water's role in the aggregation and LLPS.*

We thank the reviewer for pointing this out. We agree that we have not explored or discussed the role of water in aS aggregation or LLPS. We would like to convey that we would like to explore that in detail in a separate study altogether. However we have updated the "Discussion" section with the following lines to convey to the readers the importance water plays in aggregation and LLPS of aS.

Page 24: "The significance of the solvent in alpha-synuclein (αS) aggregation remains underexplored. Recent studies [26, 55] underscore the pivotal role of water as a solvent in LLPS. It suggests that comprehending the solvent's role, particularly water, is essential for attaining a deeper grasp of the thermodynamic and physical aspects of αS LLPS and aggregation. By delving into the solvent's contribution, researchers can uncover additional factors influencing αS aggregation. Such insights hold the potential to advance our comprehension of protein aggregation phenomena, crucial for devising strategies to address diseases linked to protein misfolding and aggregation, notably Parkinson's disease. Future investigations focusing on elucidating the interplay between αS, solvent (especially water), and other environmental elements could yield valuable insights into the mechanisms underlying LLPS and aggregation. Ultimately, this could aid in the development of therapeutic interventions or preventive measures for Parkinson's and related diseases."

> *(2) This paper discussed the effects of crowders and salt on the surface tension of the droplets.*

> *The calculation of the surface tension relies on the droplet shape. However, for the formed clusters in the MD simulations, the typical size is <10, which may be too small to rigorously define the droplet shape. As shown in previous work cited by this paper [Benayad et al., J. Chem. Theory Comput. 2021, 17, 525−537], the calculated surface tension becomes stable when the chain number is larger than 100.*

We appreciate the insightful feedback from the reviewer. However, we would like to emphasize that the αS droplets exhibit a highly liquid-like behavior, characterized by frequent exchanges of chains between the dense and dilute phases, alongside a slow aggregation process. In the study by Benayad et al. (2020, JCTC) [ref. 30], FUS-LCD was the protein of choice at concentrations in the (mM) range. FUS-LCD is known to undergo very rapid LLPS at concentrations lower than 100 (μM) where for αS the critical concentration for LLPS is 500 (μM) and undergoes slower aggregation than FUS. Moreover, the diffusion constant of αS inside newly formed droplets (no liquid to solid phase transition has occurred) has been estimated to be 0.23-0.58 μm2/s (Ray et al, 2020, Nat. Comm.). The value of diffusion constant for FUS-LCD inside LLPS droplets has been estimated to be 0.17 μm2/s (Murthy et al. 2023, Nat. Struct. and Mol. Biol.). These prove that αS forms droplets that are less viscous than that formed by FUS-LCD. This dynamic nature impedes the formation of large droplets in the simulations, making it challenging to rigorously calculate surface tension from interfacial width, which, in turn, necessitates the computation of *g(r)* between water and the droplet.

Furthermore, it's essential to note that our primary aim in calculating surface tension was not to determine its absolute value. Rather, we aimed to compare surface tensions obtained for the three distinct environments explored in this study. Hence, our primary objective is to compare the distributions of surface tensions rather than focusing solely on the mean values obtained. The distributions shown in Figure 4a clearly show a trend which we have stated in the article.

> *(3) In this work, the Martini 3 force field was modified by rescaling the LJ parameters \epsilon and \sigma with a common factor \lambda. It has not been very clearly described in the manuscript why these two different parameters can be rescaled by a common factor and why it is necessary to separately tune these two parameters, instead of just tuning the coefficient \epsilon as did in a previous work [Larsen et al., PLoS Comput Biol 16: e1007870].*

We thank the reviewer for the comment. We think that the distance of the first hydration layer also should have an impact on aggregation/LLPS. Here we are scaling both the epsilon and sigma. A higher epsilon of water-protein interactions mean higher the energy required for removal of water molecules (dehydration) when a chain goes from the dilute to the dense phase. A higher sigma on the other hand means that the hydration shell will also be at a larger distance making dehydration easier. Moreover, tuning both (either by same or different parameter) required a change of the overall protein-water interaction by only 1%, thereby requiring only considerably minimal change in forcefield parameters (compared to the case where only epsilon is being tuned which required 6-10% change in epsilon from its original values.) . Thus we think one of the ways of tuning water-protein interactions which requires minimal retuning of Martini 3 is by optimizing both epsilon and sigma. However whether a single scaling parameter is good enough requires further exploration and is outside the scope of the current study. More importantly it would introduce another free parameter into the system and the lesser the number of free parameters, the better. For this study, a single parameter sufficed as depicted in Figure 9. To inform the readers of why we chose to scale both sigma and epsilon, we have added the following in the main text:

Page 25-26: “Increasing the ϵ value of water-protein interactions results in a higher energy demand for removing water molecules (dehydration) as a chain transitions from the dilute to the dense phase. Conversely, a higher σ value implies that the hydration shell will be at a greater distance, facilitating dehydration if a chain moves into the dilute phase. Therefore, adjusting water-protein interactions based on the protein's single-chain behavior may not significantly influence the protein's phase behavior. Furthermore, fine-tuning both ϵ and σ parameters only requires a minimal change in the overall protein-water interaction (1%). As a result, this adjustment minimally alters the force field parameters."

> *(4) Both the sizes and volume fractions of the crowders can affect the protein association. It will be interesting to perform MD simulations by adding crowders with various sizes and volume fractions. In addition, in this work, the crowders were modelled by fullerenes, which contribute to protein aggregation mainly by entropic means as discussed in the manuscript. It is not very clear how the crowder effect is sensitive to the chemical nature of the crowders (e.g., inert crowders with excluded volume effect or crowders with non-specific attractive interactions with proteins, etc) and therefore the force field parameters.*

We thank the reviewer for a potential future direction. In this investigation our main focus was to simulate the inertness features of crowders only, to ensure that only entropic effect of the crowders are explored. Although this study focuses on the factors that enable aS to form an aggregates/LLPS under different environmental conditions, it would be interesting to explore in a systematic way the mechanism of action of crowders of varying shapes, sizes and interactions. Therefore we added the following lines in the "Discussion" section to let the readers know that this is also a future prospect of investigation.

Page 22: "Under physiological conditions, crowding effects emerge prominently. While crowders are commonly perceived to be inert, as has been considered in this investigation, the morphology, dimensions, and chemical interactions of crowding agents with αS in both dilute and dense phases may potentially exert considerable influence on its LLPS. Hence, a comprehensive understanding through systematic exploration is another avenue that warrants extensive investigation."

> ***Reviewer #1 (Recommendations For The Authors):***
>
> *(1) Figure S1. The title of the figure and the description in the figure caption are inconsistent?*

We thank the reviewer for the comment and we have updated the article with the correct caption.

> *(2) Page 14, line 3, the authors may want to provide more descriptions of the "ms1", "ms2", and "ms3" for better understanding.*

We are grateful to the reviewer for pointing this out. We have added a line describing in brief what "ms1", "ms2" and "ms3" represent. It reads "*Subsequent to the investigation, we utilize three representative conformations, each corresponding to one of the macrostates. We designate these macrostates as 1 (ms1), 2 (ms2), and 3 (ms3) (Figure S7)*" (Page 28)

> *(3) Page 20, the authors may want to briefly explain how the normalized Shannon entropy was calculated.*

We thank the reviewer for pointing this out. This is plain Shannon Entropy and the word "normalized" should not have been there. To avoid confusion we have provided the equation we have used to calculate the Shannon entropy (Eq 8) (Page 21).

Reviewer #2 (Public Review):

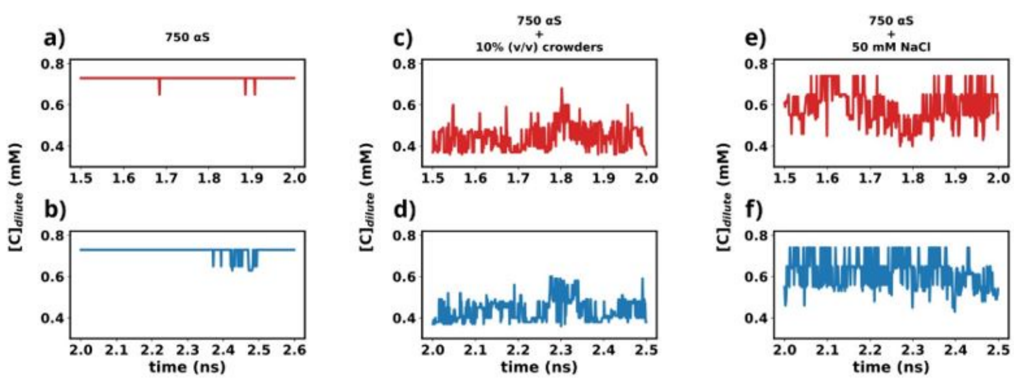
In the manuscript "Modulation of α-Synuclein Aggregation Amid Diverse Environmental Perturbation", Wasim et al describe coarse-grained molecular dynamics (cgMD) simulations of α-Synuclein (αS) at several concentrations and in the presence of molecular crowding agents or high salt. They begin by bench-marking their cgMD against all-atom simulations by Shaw. They then carry 2.4-4.3 μs cgMD simulations under the above-noted conditions and analyze the data in terms of protein structure, interaction network analysis, and extrapolated fluid mechanics properties. This is an interesting study because a molecular scale understanding of protein droplets is currently lacking, but I have a number of concerns about how it is currently executed and presented.

We thank the reviewer for finding our study interesting.

(1) It is not clear whether the simulations have reached a steady state. If they have not, it invalidates many of their analysis methods and conclusions.

We have used the last 1 μs (1.5-2.5 1 μs) from each simulation for further analysis in this study. To understand whether the simulations have reached steady state or not, we plot the time profile of the concentration of the protein in the dilute phase for all three cases.

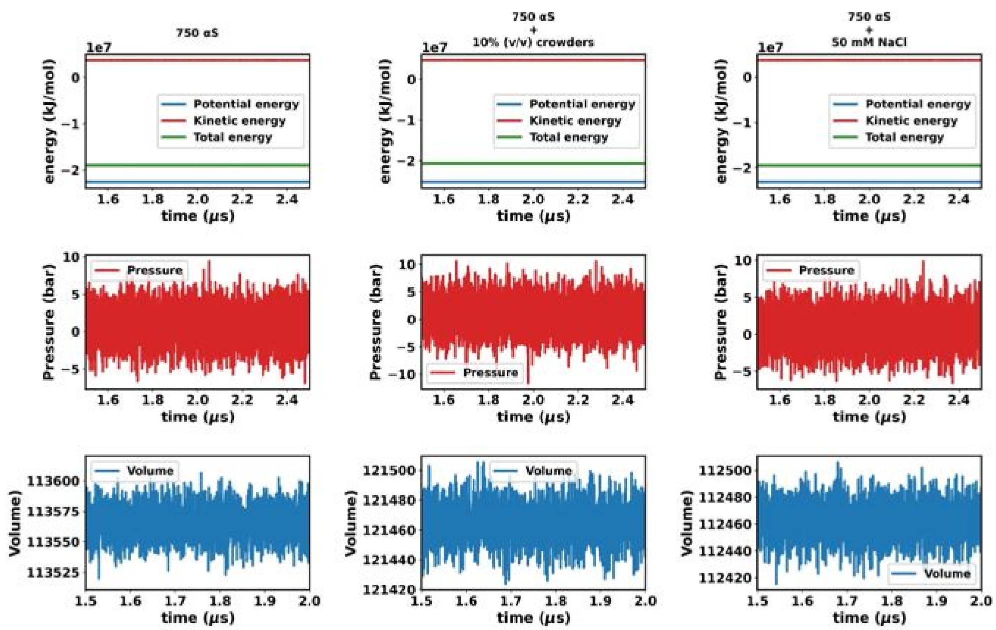
Author response image 1.



Except for the scenario of only αS (Figures a and b), the rest show very steady concentrations across various sections of the trajectory (Figures c-f). The larger sudden fluctuations observed inFigures a and b are due to the fact that only αS undergo very slow spontaneous aggregation and owing to the fact that the dense phase itself is very fluxional, addition/removal of a few chains to/from the dense to dilute phase register themselves as large fluctuations in the protein concentration in the dilute phase. For the other two scenarios (Figures c-f) aggregation has been accelerated due to the presence of crowders/salt. This causes larger aggregates to be formed. Therefore addition/removal of one or two chains does not significantly affect the concentration and we do not see such sudden large jumps. In summary, the large jumps seen in Figures a and b are due to slow, fluxional aggregation of pure αS and finite size effects. However as these still are only fluctuations, we posit that the systems have reached steady states. This claim is further supported by the following figure where the time profile of a few useful system wide macroscopic properties show no change between 1.5-2.5 μs.

We also have added a brief discussion in the Methods section (Page 29-30) with these figures in the Supplementary Information.

Author response image 2.



"In this study, we utilized the final 1 μs from each simulation for further analysis. To ascertain whether the simulations have achieved a steady state, we plotted the time profile of protein concentration in the dilute phase for all three cases. Except for minor intermittent fluctuation involving only αS in neat water (Figures S8a and S8b), the remaining cases exhibit notably stable concentrations throughout various segments of the trajectory (Figures S8 c-f). The relatively higher fluctuations observed in Figures S8a and b stem from the slow, spontaneous aggregation of αS alone, compounded by the inherently ambiguous nature of the dense phase.

Consequently, the addition or removal of a few chains from the dense to the dilute phase results in significant fluctuations in protein concentration within the dilute phase. Conversely, in the other two scenarios (Figures S8c-f), aggregation is expedited by the presence of crowders/salt, leading to the formation of larger aggregates. Consequently, the addition or removal of one or two chains has negligible impact on concentration, thereby mitigating sudden large jumps. In summary, the conspicuous jumps depicted in Figures S8a and b arise from the gradual, fluctuating aggregation of pure αS and finite size effects. However, since these remain within the realm of fluctuations, we assert that the systems have indeed reached steady states. This assertion is bolstered by the subsequent figure, where the time profile of several pertinent system-wide macroscopic properties reveals no discernible change between 1.5-2.5 μs (Figures S9)."

> *(2) The benchmarking used to validate their cgMD methods is very minimal and fails to utilize a large amount of available all-atom simulation and experimental data.*

We disagree with the reviewer on this point. We have cited multiple previous studies [26, 27] that have chosen Rg as a metric of choice for benchmarking coarse-grained model and have used a reference (experimental or otherwise) to tune Martini force fields. Majority of the notable literature where Rg was used as a benchmark during generation of new coarse-grained force fields are works by Dignon et al. (PLoS Comp. Biol.) [ref. 25], Regy et al (Protein Science. 2021) [ref. 26], Joseph et al.(Nature Computational Science. 2021) [ref. 27] and Tesei et al (Open Research Europe, 2022) [ref. 28]. From a polymer physics perspective, tuning water-protein interactions is simply changing the solvent characteristics for the biopolymer and Rg has been generally considered a suitable metric in the case of coarse-grained model. Moreover we try to match the distribution of the Rg rather than only the mean value. This suggests that at a single molecule level, the cgMD simulations at the optimum water of water-protein interactions would allow the protein to sample the conformations present in the reference ensemble. We use the extensively sampled 70 μs all-atom data from DE Shaw Research to obtain the reference Rg distribution. Also we perform a cross validation by comparing the fraction of bound states in all-atom and cgMD dimer simulations which also seem to corroborate well with each other at optimum water-protein interactions. To let the readers understand the rationale behind choosing Rg we have added a section in the Methods section (Page 25) that explains why Rg is plausibly a good metric for tuning water-protein interactions in Martini 3, at least when dealing with IDPs.

Our optimized model is further supported by the FRET experiments by Ray et al. [6]. They found that interchain NAC-NAC interactions drive LLPS. Residue level contact maps obtained from our simulations also show decreased intrachain NAC-NAC interactions with an increased interchain NAC-NAC interactions inside the droplet. This corroborates well with the experimental observations and furthermore validates the metrics we have used for optimization of the water-protein interactions. However the comparison with the FRET data by Ray et al. was not present earlier and we have added the following lines in the updated draft.

Page17: "Thus we observed that increased inter-chain NAC-NAC regions facilitate the formation of αS droplets which also have previously been seen from FRET experiments on αS LLPS

droplets[6]."

> *(3) They also miss opportunities to compare their simulations to experimental data on aSyn protein droplets.*

We thank the reviewer for pointing this out. We have tried to compare the results from our simulations to existing experimental FRET data on αS. Please see the previous response where we have described our comparison with FRET observations.

> *(4) Aspects such as network analysis are not contextualized by comparison to other protein condensed phases.*

For a proper comparison between other protein condensed phases, we would require the position phase space of such condensates which is not readily available. Therefore we tried to explain it in a simpler manner to paint a picture of how αS forms an interconnecting network inside the droplet phase.

> *(5) Data are not made available, which is an emerging standard in the field.*

We thank the reviewer for mentioning this. We have provided the trajectories between 1.5-2.5 μs, which we used for the analysis presented in the article, via a zenodo repository along with other relevant files related to the simulations (https://zenodo.org/records/10926368).

> *Firstly, it is not clear that these systems are equilibrated or at a steady state (since protein droplets are not really equilibrium systems). The authors do not present any data*

> *showing time courses that indicate the system to be reaching a steady state. This is problematic for several of their data analysis procedures, but particularly in determining free energy of transfer between the condensed and dilute phases based on partitioning.*

We have addressed this concern as stated previously in the response. We have updated the article accordingly.

> *Secondly, the benchmarking that they perform against the 73 μs all-atom simulation of aSyn monomer by Shaw and coworkers provides only very crude validation of their cgMD models based on reproducing Rg for the monomer. The authors should make more extensive comparisons to the specific conformations observed in the DE Shaw work. Shaw makes the entire trajectory publicly available. There are also a wealth of experimental data that could be used for validation with more molecular detail. See for example, NMR and FRET data used to benchmark Monte Carlo simulations of aSyn monomer (as well as extensive comparisons to the Shaw MD trajectory) in Ferrie at al: A Unified De Novo Approach for Predicting the Structures of Ordered and Disordered Proteins, J. Phys. Chem. B 124 5538-5548 (2020)*
>
> *DOI:10.1021/acs.jpcb.0c02924*
>
> *I note that NMR measurements of aSyn in liquid droplets are available from Vendruscolo: Observation of an α-synuclein liquid droplet state and its maturation into Lewy body-like assemblies, Journal of Molecular Cell Biology, Volume 13, Issue 4, April 2021, Pages 282-294, https://doi.org/10.1093/jmcb/mjaa075.*
>
> *In addition, there are FRET studies by Maji: Spectrally Resolved FRET Microscopy of α-Synuclein Phase-Separated Liquid Droplets, Methods Mol Biol 2023:2551:425-447. doi: 10.1007/978-1-0716-2597-2_27.*
>
> *So the authors are missing opportunities to better validate the simulations and place their structural understanding in greater context. This is just based on my own quick search, so I am sure that additional and possibly better experimental comparisons can be found.*

We have performed a comparison with existing FRET measurements by Ray et al. (2020) as discussed in a previous response and also updated the same in the article. The doi (10.1007/978-1-0716-2597-2_27) provided by the reviewer is however for a book on Methods to characterize protein aggregates and does not contain any information regarding the observations from FRET experiments. The other doi (https://doi.org/10.1093/jmcb/mjaa075) for the article from Vendrusculo group does not contain information directly relevant to this study. Moreover NMR measurements cannot be predicted from cgMD since full atomic resolution is lost upon coarse-graining of the protein . A past literature survey by the authors found very little scientific literature on molecular level characterization of αS LLPS droplets.

> *Thirdly, the small word network analysis is interesting, but hard to contextualize. For instance, the 8 Å cutoff used seems arbitrary. How does changing the cutoff affect the value of S determined? Also, how does the value of S compare to other condensed phases like crystal packing or amyloid forms of aSyn?*

The 8 Å cutoff is actually arbitrary since a distance based clustering always requires a cutoff which is empirically decided. However 8 Å is quite large compared to other cutoffs used for distance based clustering. For example in ref 26, 5 Å was used as a cutoff for calculation of protein clusters. Larger cutoffs will lead to sparser network structures. However we used the same cutoff for all distance based clustering which makes the networks obtained comparable.

We wanted to perform a comparison among the networks formed by αS under different environmental conditions.

> *Fourthly, I see no statement on data availability. The emerging standard in the computational field is to make all data publicly available through Github or some similar mechanism.*

We thank the reviewer for pointing this out and we have provided the raw data between 1.5-2.5 μs for each scenario along with other relevant files via a zenodo repository (https://zenodo.org/records/10926368).

> *Finally, on page 16, they discuss the interactions of aSyn(95-110), but the sequence that they give is too long (seeming to contain repeated characters, but also not accurate). aSyn(95-110) = VKKDQLGKNEEGAPQE. Presumably this is just a typo, but potentially raises concerns about the simulations (since without available data, one cannot check that the sequence is accurate) and data analysis elsewhere.*

This indeed is a typographical error. We have updated the article with the correct sequence. The validity of the simulations can be verified from the data we have shared via the zenodo repository (https://zenodo.org/records/10926368).

https://doi.org/10.7554/eLife.95180.2.sa0